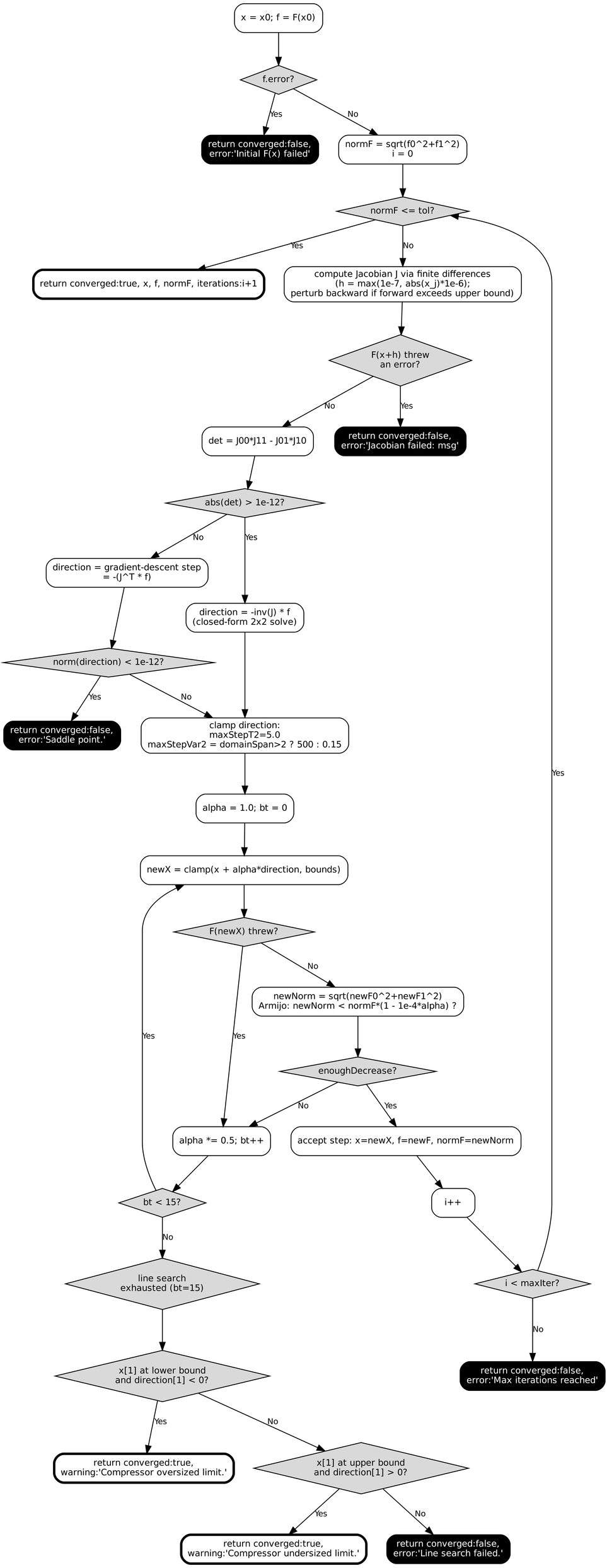
x = x0; f = F(x0)
f.error?
Yes
No
return converged:false, error:'Initial F(x) failed'
normF = sqrt(f0^2+f1^2)
i = 0
normF <= tol?
Yes
No
return converged:true, x, f, normF, iterations:i+1
compute Jacobian J via finite differences
(h = max(1e-7, abs(x_j)*1e-6);
perturb backward if forward exceeds upper bound)
F(x+h) threw an error?
No
Yes
det = J00*J11 - J01*J10
return converged:false, error:'Jacobian failed: msg'
abs(det) > 1e-12?
No
Yes
direction = gradient-descent step = -(J^T * f)
direction = -inv(J) * f
(closed-form 2x2 solve)
norm(direction) < 1e-12?
Yes
No
return converged:false, error:'Saddle point.'
clamp direction:
maxStepT2=5.0
maxStepVar2 = domainSpan>2 ? 500 : 0.15
alpha = 1.0; bt = 0
newX = clamp(x + alpha*direction, bounds)
F(newX) threw?
Yes
No
newNorm = sqrt(newF0^2+newF1^2)
Armijo: newNorm < normF*(1 - 1e-4*alpha) ?
Yes
enoughDecrease?
No
Yes
alpha *= 0.5; bt++
accept step: x=newX, f=newF, normF=newNorm
bt < 15?
Yes
No
i++
line search exhausted (bt=15)
i < maxIter?
Yes
No
return converged:false, error:'Max iterations reached'
x[1] at lower bound and direction[1] < 0?
Yes
No
return converged:true, warning:'Compressor oversized limit.'
x[1] at upper bound and direction[1] > 0?
Yes
No
return converged:true, warning:'Compressor undersized limit.'
return converged:false, error:'Line search failed.'